

GEMM and MLP Layer

The MLP layer can be broken down into the following steps:

$$1 \ G = XW_g \text{ 2048 x 4096, 4096 x 14336}$$

$$2 \ U = XW_u \text{ 2048 x 4096, 4096 x 14336}$$

$$3 \ Z = \text{SiLU}(G) \otimes U, \text{ 2048 x 14336, 2048 x 14336}$$

$$4 \ Y = ZW_d \text{ 2048 x 14336, 14336 x 4096}$$

5 Residual

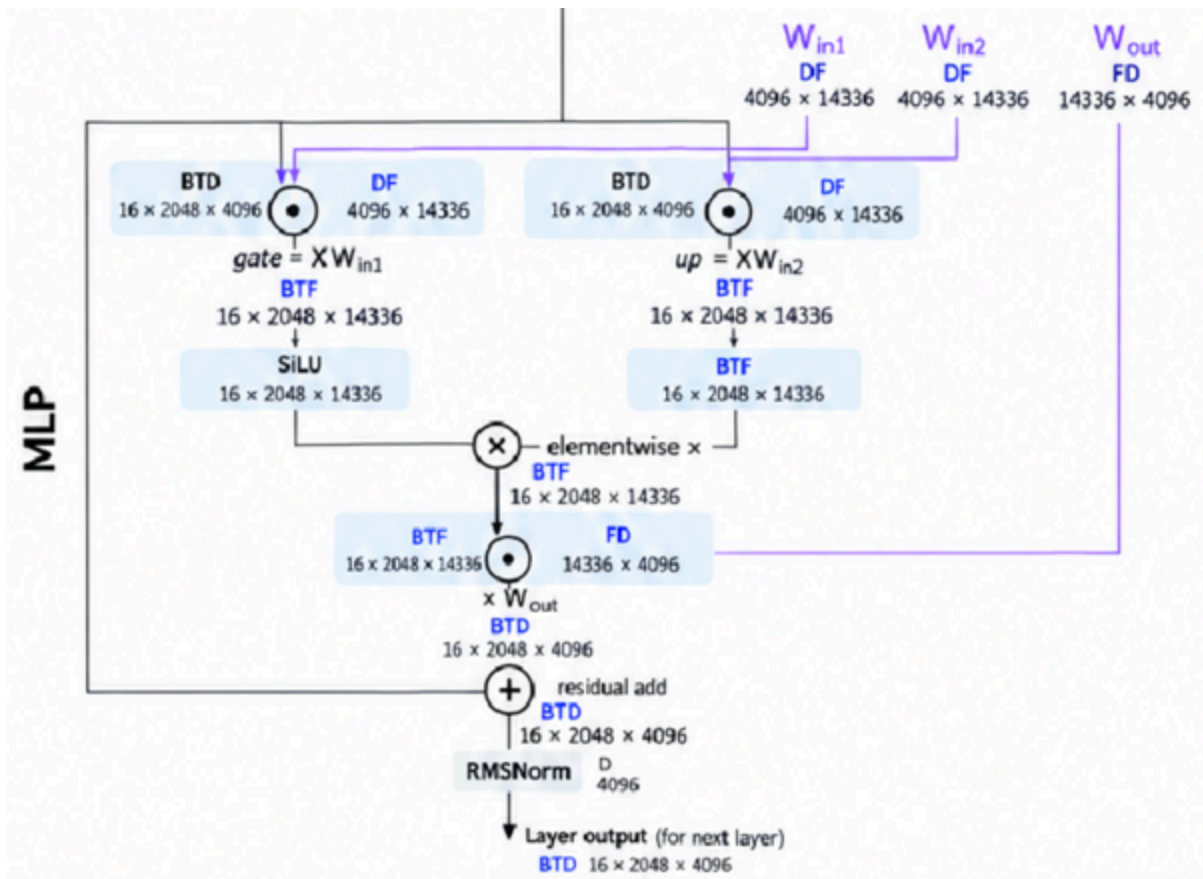
6 RMSNorm

MLP is tensor-operation intense, most of the on-chip happens on GEMM operations. The strategy for parallelize GEMM on SA1/SA2 uses the same idea in FA2: maximize the intensity of SA unit and non-GEMM operations run on the VCPU concurrently.

Ref: Pope et al., "Efficiently Scaling Transformer Inference" (arXiv:2211.05102)

```
In [1]: import sys
        if "code" not in sys.path: sys.path.insert(0, "code") # model modules live in
        code/
```

```
In [2]: from helper import plot
        from mlp import *
        plot("img/MLP.png", 600)
```



1. MLP in Prefill

Tiling is essential for the MLP, as the following simple calculation shows:

```
In [3]: # a whole weight matrix can not stay resident in SRAM
from helper import D, F
from tile_model import Hardware

sram = Hardware().sram_bytes
for name, rows, cols in [("Wg", D, F), ("Wu", D, F), ("Wd", F, D)]:
    b = rows * cols # INT8 weights, 1 byte each
    print(f"{name}: {b/2**20:5.1f} MiB fits in {sram/2**20:.0f} MiB SRAM? {b <= sram}")
```

```
Wg: 56.0 MiB fits in 16 MiB SRAM? False
Wu: 56.0 MiB fits in 16 MiB SRAM? False
Wd: 56.0 MiB fits in 16 MiB SRAM? False
```

Following the same setting as in FA2, all GEMM operations are carried out on the SA, while the VCPU handles the residual and RMSNorm parts.

3 For loops:

1 loop through sequence length T

2 loop through feature F

3 loop through reduction D

The tile size therefore: $M_t \times R @ R \times F_t$ for SA

```
FOR each token tile t (Mt rows)

    DMA load X16 tile                                # prefetched one feature
tile early
    VCPU quantize X16 -> X8 operands                # static SmoothQuant
scale, one pass

    FOR each hidden-feature tile f (Ft columns)

        DMA load Wg|Wu(f)                            # slot reused after
gate+up(f-2)
        DMA load Wd(f)                                # slot reused after
down(f-2)

        # ---- gate + up: [Mt x 2Ft] = X8[Mt x D] . Wg|Wu[D x 2Ft]
FOR each reduction chunk k across D (R = 256)
        SA1/SA2 compute X8 . Wg|Wu[k] partial products
        SRAM write raw INT16 partials
        VCPU G U 32 partials                        # overlaps the next chunk
VCPU last chunk: apply scales, Z = SiLU(G) * U,
        quantize -> Z8 operands                    # fused into the final
accumulate

        # ---- down: [Mt x D] += Z8[Mt x Ft] . Wd[Ft x D]
FOR each reduction chunk k across Ft
        SA1/SA2 compute Z8 . Wd[k] partial products
        SRAM write raw INT16 partials
        VCPU Y32 (|=|+=) partials                  # queued after
gate+up(f+2) work,
                                                    # first chunk of f=0
writes Y32
    END FOR

    DMA load residual X16 tile
    VCPU Y16 = Y32 + X16, output scale
    DMA store Y16                                    # overlaps the next token
tile
END FOR
```

In [4]: *# Mt = 64 on SA1 and 128 on SA2*

In [5]: *# The parameter setting identical with FA2, A row split of 192 and R = 256 to maximize the performance*

hw = Hardware()

```
mlp(M=2048, D=4096, F=14336, Mt=192, Ft=256, Mt_sa1=None, R=256,  
    residual=True, hw=hw, overlap=True, verbose=True)["time_ms"]
```

x0 load X16	DMA	@	0	24,976 cycles
24.976 us SRAM 1536.00 KiB				
x0 quantize X -> X8 operands	VCPU	@	24,976	18,732 cycles
18.732 us SRAM 4370.00 KiB				
x0 f0 load Wg Wu	DMA	@	24,976	33,168 cycles
33.168 us SRAM 4370.00 KiB				
x0 f0 gate+up k0	ARRAYS	@	58,144	45,796 cycles
45.796 us SRAM 5092.00 KiB				
x0 f256 load Wg Wu	DMA	@	58,144	33,168 cycles
33.168 us SRAM 5092.00 KiB				
x0 f256 load Wd	DMA	@	91,312	16,784 cycles
16.784 us SRAM 6116.00 KiB				
x0 f0 G,U32 = k0	VCPU	@	103,940	4,908 cycles
4.908 us SRAM 6692.00 KiB				
x0 f0 gate+up k256	ARRAYS	@	103,940	45,796 cycles
45.796 us SRAM 6692.00 KiB				
x0 f0 load Wd	DMA	@	108,096	16,784 cycles
16.784 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k256	VCPU	@	149,736	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k512	ARRAYS	@	149,736	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k512	VCPU	@	195,532	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k768	ARRAYS	@	195,532	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k768	VCPU	@	241,328	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k1024	ARRAYS	@	241,328	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k1024	VCPU	@	287,124	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k1280	ARRAYS	@	287,124	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k1280	VCPU	@	332,920	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k1536	ARRAYS	@	332,920	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k1536	VCPU	@	378,716	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k1792	ARRAYS	@	378,716	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k1792	VCPU	@	424,512	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k2048	ARRAYS	@	424,512	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k2048	VCPU	@	470,308	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k2304	ARRAYS	@	470,308	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k2304	VCPU	@	516,104	7,980 cycles
7.980 us SRAM 7716.00 KiB				
x0 f0 gate+up k2560	ARRAYS	@	516,104	45,796 cycles
45.796 us SRAM 7716.00 KiB				
x0 f0 G,U32 += k2560	VCPU	@	561,900	7,980 cycles
7.980 us SRAM 7716.00 KiB				

x0 f0 gate+up k2816	ARRAYS @	561,900	45,796 cycles
45.796 us SRAM 7716.00 KiB			
x0 f0 G,U32 += k2816	VCPU @	607,696	7,980 cycles
7.980 us SRAM 7716.00 KiB			
x0 f0 gate+up k3072	ARRAYS @	607,696	45,796 cycles
45.796 us SRAM 7716.00 KiB			
x0 f0 G,U32 += k3072	VCPU @	653,492	7,980 cycles
7.980 us SRAM 7716.00 KiB			
x0 f0 gate+up k3328	ARRAYS @	653,492	45,796 cycles
45.796 us SRAM 7716.00 KiB			
x0 f0 G,U32 += k3328	VCPU @	699,288	7,980 cycles
7.980 us SRAM 7716.00 KiB			
x0 f0 gate+up k3584	ARRAYS @	699,288	45,796 cycles
45.796 us SRAM 7716.00 KiB			
x0 f0 G,U32 += k3584	VCPU @	745,084	7,980 cycles
7.980 us SRAM 7716.00 KiB			
x0 f0 gate+up k3840	ARRAYS @	745,084	45,796 cycles
45.796 us SRAM 7716.00 KiB			
x0 f0 G,U -> SiLU(G)*U -> Z8	VCPU @	790,880	5,308 cycles
5.308 us SRAM 7782.00 KiB			
x0 f256 gate+up k0	ARRAYS @	790,880	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f512 load Wg Wu	DMA @	790,880	33,168 cycles
33.168 us SRAM 7782.00 KiB			
x0 f256 G,U32 = k0	VCPU @	836,676	4,908 cycles
4.908 us SRAM 7782.00 KiB			
x0 f256 gate+up k256	ARRAYS @	836,676	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k256	VCPU @	882,472	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k512	ARRAYS @	882,472	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k512	VCPU @	928,268	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k768	ARRAYS @	928,268	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k768	VCPU @	974,064	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k1024	ARRAYS @	974,064	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k1024	VCPU @	1,019,860	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k1280	ARRAYS @	1,019,860	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k1280	VCPU @	1,065,656	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k1536	ARRAYS @	1,065,656	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k1536	VCPU @	1,111,452	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k1792	ARRAYS @	1,111,452	45,796 cycles
45.796 us SRAM 7782.00 KiB			
x0 f256 G,U32 += k1792	VCPU @	1,157,248	7,980 cycles
7.980 us SRAM 7782.00 KiB			
x0 f256 gate+up k2048	ARRAYS @	1,157,248	45,796 cycles
45.796 us SRAM 7782.00 KiB			

x0 f256 G,U32 += k2048	VCPU	@	1,203,044	7,980 cycles
7.980 us SRAM 7782.00 KiB				
x0 f256 gate+up k2304	ARRAYS	@	1,203,044	45,796 cycles
45.796 us SRAM 7782.00 KiB				
x0 f256 G,U32 += k2304	VCPU	@	1,248,840	7,980 cycles
7.980 us SRAM 7782.00 KiB				
x0 f256 gate+up k2560	ARRAYS	@	1,248,840	45,796 cycles
45.796 us SRAM 7782.00 KiB				
... [22,157 lines of verbose trace omitted in PDF; full output in 4.MLP.html] ...				
x1920 f14080 G,U32 += k3072	VCPU	@	660,161,740	5,548 cycles
5.548 us SRAM 9462.00 KiB				
x1920 f14080 gate+up k3328	ARRAYS	@	660,161,740	34,404 cycles
34.404 us SRAM 9462.00 KiB				
x1920 f14080 G,U32 += k3328	VCPU	@	660,196,144	5,548 cycles
5.548 us SRAM 9462.00 KiB				
x1920 f14080 gate+up k3584	ARRAYS	@	660,196,144	34,404 cycles
34.404 us SRAM 9462.00 KiB				
x1920 f14080 G,U32 += k3584	VCPU	@	660,230,548	5,548 cycles
5.548 us SRAM 9462.00 KiB				
x1920 f14080 gate+up k3840	ARRAYS	@	660,230,548	34,404 cycles
34.404 us SRAM 9462.00 KiB				
x1920 f14080 G,U -> SiLU(G)*U -> Z8	VCPU	@	660,264,952	3,772 cycles
3.772 us SRAM 8454.00 KiB				
x1920 f13824 down k0	ARRAYS	@	660,264,952	273,636 cycles
273.636 us SRAM 8454.00 KiB				
x1920 f13568 Y32 += k0	VCPU	@	660,268,724	42,412 cycles
42.412 us SRAM 8054.00 KiB				
x1920 f14080 down k0	ARRAYS	@	660,538,588	273,636 cycles
273.636 us SRAM 6980.00 KiB				
x1920 f13824 Y32 += k0	VCPU	@	660,538,588	42,412 cycles
42.412 us SRAM 6980.00 KiB				
x1920 f14080 Y32 += k0	VCPU	@	660,812,224	42,412 cycles
42.412 us SRAM 4754.00 KiB				
x1920 encode Y16 + residual	VCPU	@	660,854,636	33,069 cycles
33.069 us SRAM 4608.00 KiB				
x1920 store Y16	DMA	@	660,887,705	16,734 cycles
16.734 us SRAM 1024.00 KiB				

Total: 660.904439 ms (660,904,439 cycles), 546 MAC/cycle

Busy: DMA 4.8%, VCPU 16.7%, SA1+SA2 100.0%

Weights streamed: 1,848 MiB (11 token tiles)

Peak SRAM: 13,492,224 bytes (80.420%)

Out[5]: 660.9044389999999

2. MLP Decoder

MLP in decoder has 1 sequence length (Same situation in FA GEMM), thus, we apply the same Transpose trick on the SA input to fulfill the rows:

$$1 \text{ G} = XW_g \text{ } 16 \times 1 \times 4096, 4096 \times 14336$$

$$2 \text{ U} = XW_u \text{ } 16 \times 1 \times 4096, 4096 \times 14336$$

$$3\ Z = \text{SiLU}(G) \otimes U, 16 \times 1 \times 14336, 1 \times 14336$$

$$4\ Y = ZW_d\ 16 \times 1 \times 14336, 14336 \times 4096$$

The mapping is transposed (features on the array rows, the M tokens on the 16-wide output axis) because 16 tokens cannot fill the 48 rows of a job pass.

```
# X16: [M, D]          M = decode batch, one new token per
sequence
# Wg8|Wu8^T: [2F, D]   INT8, per-row scales, pre-packed in array
operand
# Wd8^T:      [D, F]    layout in DRAM, TRANSPOSED w.r.t. prefill

DMA  load X16                      # the whole step: M rows, a
few hundred KiB
VCPU quantize X16 -> X8^T operands  # [D, M], static SmoothQuant
scale

FOR each hidden-feature tile f (Ft features)

    DMA  load Wg|Wu^T(f)            # [2Ft, D], slot reused
after gate+up(f-2)
    DMA  load Wd^T(f)              # [D, Ft], slot reused
after down(f-2)

    # ---- gate + up: [2Ft x M] = Wg|Wu^T[2Ft x D] . X8^T[D x M]
    FOR each reduction chunk k across D (R = 256)
        _**SA1/SA2 compute Wg|Wu^T[k] . X8^T[k] partial products**_
        SRAM write raw INT16 partials
        VCPU G32 U32 partials      # overlaps the next chunk
    VCPU last chunk: apply scales, Z = SiLU(G) * U,
        quantize -> Z8^T [Ft x M]  # fused into the final
accumulate

    # ---- down: [D x M] += Wd^T[D x Ft] . Z8^T[Ft x M]
    FOR each reduction chunk k across Ft
        SA1/SA2 compute Wd^T[k] . Z8^T[k] partial products
        SRAM write raw INT16 partials
        VCPU Y32 partials          # queued after gate+up(f+2) work
    END FOR

DMA  load residual X16
VCPU Y16 = Y32 + X16, output scale
DMA  store Y16
```

Finally put all the component into the performance model:

```
In [6]: mlp_decode(M=1, D=4096, F=14336, Ft=256, R=256,
                residual=True, hw=hw, overlap=True, verbose=True)["time_ms"]
```

```

load X16
0.528 us  SRAM      8.00 KiB
quantize X -> X8^T operands
0.396 us  SRAM    2078.00 KiB
f0 load Wg|Wu^T
33.168 us  SRAM    2078.00 KiB
f0 gate+up k0
4.144 us  SRAM    4152.50 KiB
f256 load Wg|Wu^T
33.168 us  SRAM    4152.50 KiB
f0 G,U32 = k0
0.448 us  SRAM    4171.00 KiB
f0 gate+up k256
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k256
0.464 us  SRAM    4171.00 KiB
f0 gate+up k512
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k512
0.464 us  SRAM    4171.00 KiB
f0 gate+up k768
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k768
0.464 us  SRAM    4171.00 KiB
f0 gate+up k1024
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k1024
0.464 us  SRAM    4171.00 KiB
f0 gate+up k1280
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k1280
0.464 us  SRAM    4171.00 KiB
f0 gate+up k1536
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k1536
0.464 us  SRAM    4171.00 KiB
f0 gate+up k1792
4.144 us  SRAM    4171.00 KiB
f0 G,U32 += k1792
0.464 us  SRAM    4171.00 KiB
f0 gate+up k2048
4.144 us  SRAM    4171.00 KiB
f256 load Wd^T
16.784 us  SRAM    5195.00 KiB
f0 G,U32 += k2048
0.464 us  SRAM    5195.00 KiB
f0 gate+up k2304
4.144 us  SRAM    5195.00 KiB
f0 G,U32 += k2304
0.464 us  SRAM    5195.00 KiB
f0 gate+up k2560
4.144 us  SRAM    5195.00 KiB
f0 G,U32 += k2560
0.464 us  SRAM    5195.00 KiB
f0 gate+up k2816
4.144 us  SRAM    5195.00 KiB

```

```

DMA      @           0      528 cycles
VCPU     @          528      396 cycles
DMA      @          528    33,168 cycles
ARRAYS   @        33,696     4,144 cycles
DMA      @        33,696    33,168 cycles
VCPU     @        37,840     448 cycles
ARRAYS   @        37,840     4,144 cycles
VCPU     @        41,984     464 cycles
ARRAYS   @        41,984     4,144 cycles
VCPU     @        46,128     464 cycles
ARRAYS   @        46,128     4,144 cycles
VCPU     @        50,272     464 cycles
ARRAYS   @        50,272     4,144 cycles
VCPU     @        54,416     464 cycles
ARRAYS   @        54,416     4,144 cycles
VCPU     @        58,560     464 cycles
ARRAYS   @        58,560     4,144 cycles
VCPU     @        62,704     464 cycles
ARRAYS   @        62,704     4,144 cycles
VCPU     @        66,848     464 cycles
ARRAYS   @        66,848     4,144 cycles
DMA      @        66,864    16,784 cycles
VCPU     @        70,992     464 cycles
ARRAYS   @        70,992     4,144 cycles
VCPU     @        75,136     464 cycles
ARRAYS   @        75,136     4,144 cycles
VCPU     @        79,280     464 cycles
ARRAYS   @        79,280     4,144 cycles

```

f0 G,U32 += k2816
 0.464 us SRAM 5195.00 KiB
 f0 gate+up k3072
 4.144 us SRAM 5195.00 KiB
 f0 load Wd^T
 16.784 us SRAM 6219.00 KiB
 f0 G,U32 += k3072
 0.464 us SRAM 6219.00 KiB
 f0 gate+up k3328
 4.144 us SRAM 6219.00 KiB
 f0 G,U32 += k3328
 0.464 us SRAM 6219.00 KiB
 f0 gate+up k3584
 4.144 us SRAM 6219.00 KiB
 f0 G,U32 += k3584
 0.464 us SRAM 6219.00 KiB
 f0 gate+up k3840
 4.144 us SRAM 6219.00 KiB
 f0 G,U -> SiLU(G)*U -> Z8^T
 0.466 us SRAM 4171.25 KiB
 f256 gate+up k0
 4.144 us SRAM 4171.25 KiB
 f512 load Wg|Wu^T
 33.168 us SRAM 6237.25 KiB
 f256 G,U32 = k0
 0.448 us SRAM 6237.25 KiB
 f256 gate+up k256
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k256
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k512
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k512
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k768
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k768
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k1024
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k1024
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k1280
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k1280
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k1536
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k1536
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k1792
 4.144 us SRAM 6237.25 KiB
 f256 G,U32 += k1792
 0.464 us SRAM 6237.25 KiB
 f256 gate+up k2048
 4.144 us SRAM 6237.25 KiB

VCPU	@	83,424	464 cycles
ARRAYS	@	83,424	4,144 cycles
DMA	@	83,648	16,784 cycles
VCPU	@	87,568	464 cycles
ARRAYS	@	87,568	4,144 cycles
VCPU	@	91,712	464 cycles
ARRAYS	@	91,712	4,144 cycles
VCPU	@	95,856	464 cycles
ARRAYS	@	95,856	4,144 cycles
VCPU	@	100,000	466 cycles
ARRAYS	@	100,000	4,144 cycles
DMA	@	100,432	33,168 cycles
VCPU	@	104,144	448 cycles
ARRAYS	@	104,144	4,144 cycles
VCPU	@	108,288	464 cycles
ARRAYS	@	108,288	4,144 cycles
VCPU	@	112,432	464 cycles
ARRAYS	@	112,432	4,144 cycles
VCPU	@	116,576	464 cycles
ARRAYS	@	116,576	4,144 cycles
VCPU	@	120,720	464 cycles
ARRAYS	@	120,720	4,144 cycles
VCPU	@	124,864	464 cycles
ARRAYS	@	124,864	4,144 cycles
VCPU	@	129,008	464 cycles
ARRAYS	@	129,008	4,144 cycles
VCPU	@	133,152	464 cycles
ARRAYS	@	133,152	4,144 cycles

f256 G,U32 += k2048	VCPU	@	137,296	464 cycles
0.464 us SRAM 6237.25 KiB				
f256 gate+up k2304	ARRAYS	@	137,296	4,144 cycles
4.144 us SRAM 6237.25 KiB				
f256 G,U32 += k2304	VCPU	@	141,440	464 cycles
0.464 us SRAM 6237.25 KiB				
f256 gate+up k2560	ARRAYS	@	141,440	4,144 cycles
4.144 us SRAM 6237.25 KiB				
... [1,947 lines of verbose trace omitted in PDF; full output in 4.MLP.html] ...				
f14080 G,U32 += k3072	VCPU	@	5,391,008	464 cycles
0.464 us SRAM 4341.25 KiB				
f14080 gate+up k3328	ARRAYS	@	5,391,008	4,144 cycles
4.144 us SRAM 4341.25 KiB				
f14080 G,U32 += k3328	VCPU	@	5,395,152	464 cycles
0.464 us SRAM 4341.25 KiB				
f14080 gate+up k3584	ARRAYS	@	5,395,152	4,144 cycles
4.144 us SRAM 4341.25 KiB				
f14080 G,U32 += k3584	VCPU	@	5,399,296	464 cycles
0.464 us SRAM 4341.25 KiB				
f14080 gate+up k3840	ARRAYS	@	5,399,296	4,144 cycles
4.144 us SRAM 4341.25 KiB				
f14080 G,U -> SiLU(G)*U -> Z8^T	VCPU	@	5,403,440	466 cycles
0.466 us SRAM 2405.00 KiB				
f13824 down k0	ARRAYS	@	5,403,440	30,680 cycles
30.680 us SRAM 2405.00 KiB				
f13568 Y32 += k0	VCPU	@	5,403,906	1,708 cycles
1.708 us SRAM 2386.50 KiB				
f14080 down k0	ARRAYS	@	5,434,120	30,680 cycles
30.680 us SRAM 1344.25 KiB				
f13824 Y32 += k0	VCPU	@	5,434,120	1,708 cycles
1.708 us SRAM 1344.25 KiB				
f14080 Y32 += k0	VCPU	@	5,464,800	1,708 cycles
1.708 us SRAM 174.00 KiB				
encode Y16 + residual	VCPU	@	5,466,508	557 cycles
0.557 us SRAM 36.00 KiB				
store Y16	DMA	@	5,467,065	478 cycles
0.478 us SRAM 8.00 KiB				

Total: 5.467543 ms (5,467,543 cycles), 32 MAC/cycle, 5467.5 us/token/layer

Busy: DMA 51.2%, VCPU 9.4%, SA1+SA2 99.3%

Weights streamed: 168 MiB per step -> DMA floor 2.753 ms

Peak SRAM: 6,552,832 bytes (39.058%)

Out[6]: 5.467543